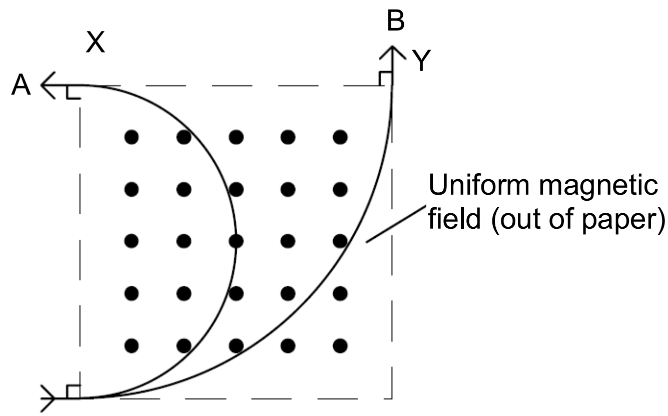
24	Particles *A* and *B* moving at the same speed enter a square region of uniform magnetic field as shown in the figure below. Particle *A* leaves at *X* while particle *B* leaves at *Y*.  If the ratio of charge to mass of particle *A* is k, what is that of particle *B*?

	A	$\frac{k}{2}$
	B	$\frac{k}{4}$
	C	$2k$
	D	$4k$

Ans: A

For A:

$Bqv = \frac{mv^2}{r} \quad \Rightarrow r = \frac{mv}{Bq} = \frac{v}{Bk}$

For B:

Radius = 2r $\Rightarrow k' = \frac{1}{2}k$